

Figure 3 — Weighted Gene Co-Expression Network Analysis (WGCNA)

Topological module identification and trait correlations in *Mycobacterium marinum* OSD-528

A — Co-Expression Network Modules

- MEturquoise: Biofilm & GPL Synthesis** ($n = 1$)
Key hubs: *mps1* ($k = 11.8$), *mps2*, *fmt*, *fadD28*. Drives PDM.
- MEblue: Mycolic Acid Cell Envelope** ($n = 37$ genes)
Key hubs: *kasA* ($k = 15.2$), *kasB*, *acpM*, *inhA*, *fbpA* (Ag85A).
- MEbrown: Type VII Secretion System** ($n = 19$)
Key hubs: *esxA* (ESAT-6, $k = 24.1$), *esxB* (CFP-10), *eccA1-eccA5*.
- MEyellow: DosR Dormancy & Antioxidant** ($n = 1$)
Key hubs: *dosR*, *dosS*, *hspX* ($k = 18.4$), *tgsl*, *katG*, *sodA*, *sigH*.
- MEgreen: Simulator-Specific Kinematics** ($n = 1$)
Key hubs: *dnaK*, *clpP1* (Clinostat) vs *cydA*, *icl1* (RPM 2.0).

B — Module-Trait Relationships

	Microgravity	Clinostat vs RPM
MEturquoise	$r = 0.99$	$r = 0.12$
MEblue	$r = 0.98$	$r = -0.16$
MEbrown	$r = 0.98$	$r = 0.04$
MEyellow	$r = 0.99$	$r = 0.10$
MEgreen	$r = 0.98$	$r = -0.01$